

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (Civil) (Credit System) Degree Examinations**

January 2023

20CV504 – TRANSPORTATION ENGINEERING - I

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Assume any missing data as per IRC.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|--------------|------------|------------|------------|
| 1. a) List and explain the various studies carried out for planning survey. | 10 | L*2 | 1 | 1 |
| b) Define overtaking sight distance. Calculate the safe overtaking sight distance for a design speed of 80 kmph on a two way traffic road. Assume that the acceleration of overtaking vehicle is 0.72 m/sec ² . Also,
(i) Compute the minimum and maximum length of overtaking zone.
(ii) Draw a neat sketch of the overtaking zone and show the position of sign post. | 10 | L3 | 2 | 2 |
| 2. a) What is saturation system of road planning? The area of certain district of India is 3,08,000 sq.km and there are 276 towns as per 1981 census. Determine the lengths of different categories of roads to be provided in this district as per 3 rd 20 year RDP. | 10 | L2 | 1 | 2 |
| b) List and explain the four surface characteristics of the pavement. | 10 | L2 | 2 | 1 |
| 3. a) Explain the steps involved in a new highway project work. | 10 | L2 | 1 | 1 |
| b) A rural road is passing through a hilly terrain in a heavy rainfall area has a horizontal curve of radius 250m. Adopt the design speed as 50 kmph, length of wheel base of the largest truck as 6.0 m and the width of pavement as 7.0 m.
Compute the following geometric design elements as per IRC:
(i) Superelevation (ii) Extra widening of pavement
(iii) Length of transition curve | 10 | L3 | 2 | 2 |

Unit – II

- | | | | | |
|---|----|----|---|---|
| 4. a) Explain the concept of PCU and enumerate any six factors which affects the PCU values. Also, indicate any two IRC recommended PCU values for different classes of vehicles on roads in rural areas. | 10 | L2 | 3 | 1 |
| b) With the help of neat sketches explain the functioning of flexible and rigid pavements. | 10 | L2 | 4 | 1 |
| 5. a) What do you mean by "bitumen emulsion" and "cutback bitumen"? Enumerate any six desirable properties of bitumen. | 10 | L2 | 3 | 1 |
| b) Estimate the annual cost of a stretch of highway from the following particulars. | | | | |

Item	Total cost Rs. In lakhs	Estimated life, years	Rate of interest %
Land	12	100	6
Earth work	9.0	40	8
Bridges, culverts and drainage	7.5	60	8
Pavement	14	15	10

Average cost of maintenance of the road is 1.5 lakhs per year.

10 L3 4 2

6. a) Define CBR value of the soil and determine the CBR value of the soil using graphical method for the following test results. Least count of the dial gauge is 0.01mm.

Penetration of plunger (division)	Load (kg)
0	0
50	5
100	16.2
150	28.1
200	40
250	48.5
300	56.5
400	67.5
500	75.2
750	89
1000	99.5
1250	106.5

- b) Explain the five factors to be considered for the performance and design of flexible pavements.

10 L3 3 2
10 L2 4 1

Unit – III

7. a) Explain with sketches, how the subsurface drainage system is provided to lower the water table. Also describe the methods of provision of granular capillary cut off to control capillary rise of ground water with sketches.

10 L2 5 1

- b) Briefly explain the following with respect to pavement:
(i) Mud Pumping (ii) Scaling (iii) Raveling (iv) Corrugation
(v) Rutting

10 L2 5 1

8. a) Briefly explain any five methods of maintenance works carried out for highways.

10 L2 5 1

- b) Explain any two methods of functional evaluation of pavements.

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
